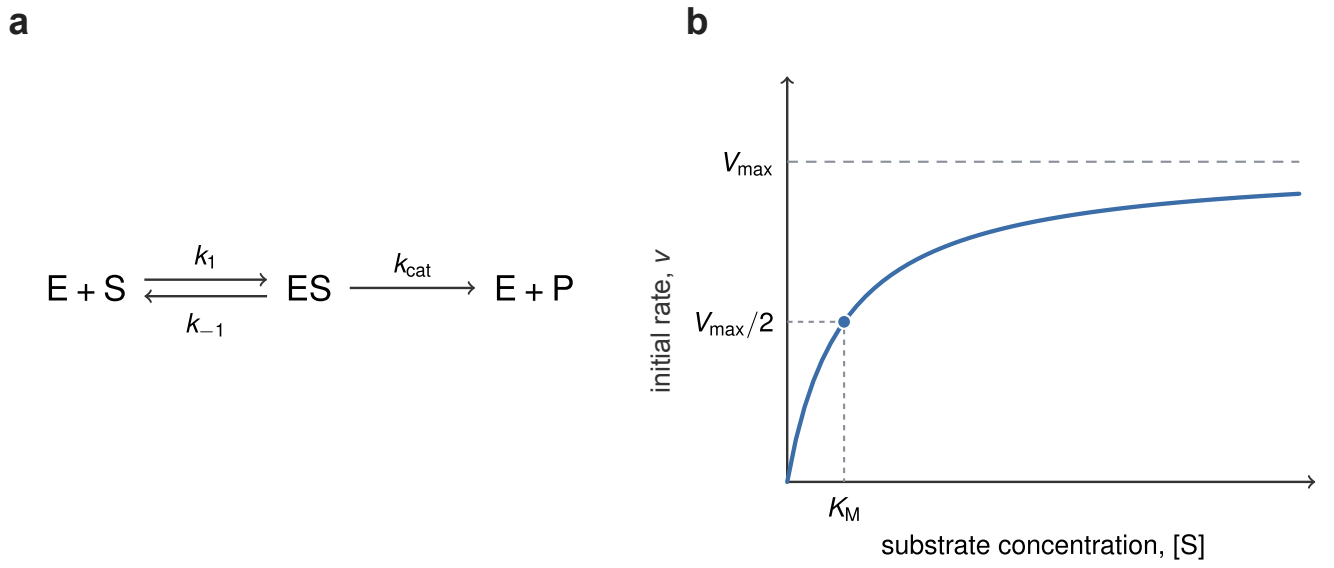

a
E + S
k_1
k_{-1}
ES
k_{cat}
E + P
b
V_{max}
$V_{max}/2$
initial rate, v
K_M
substrate concentration, [S]

a

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_{cat}} E + P$$

b

